Market Deformation Geometry

A Teichmüller Framework for Structural Change Detection
in Multi-Asset, Multi-Venue Markets

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Abstract

This paper proposes *Market Deformation Geometry*: a structure-first Teichmüller framework that measures structural change in multi-asset, multi-venue financial markets as quasiconformal deformation of synchronized price observations. The observable market is represented not as a collection of isolated price charts, but as a coupled multidimensional geometric object. Foreign exchange serves as the running example throughout, because it supplies many algebraically related instruments and multiple observable price feeds; the construction itself applies to any universe of instruments observed through several venues.

The fundamental data structure is the synchronized tensor $P(t, a, b)$, where t denotes time, a an asset — a currency pair, a metal, an index, or any other instrument — and b a broker or price-feed observation. The temporal coordinate describes the evolution of the market, the asset coordinate describes cross-asset relationships, and the broker coordinate describes different realizations of nominally identical market instruments.

The central observation is that these dimensions must not be treated as interchangeable. Connecting observations along the time axis produces a conventional price path. Connecting observations along the broker axis instead produces a cross-sectional observation geometry. Connecting assets produces a third structural direction. The paper develops finite-difference analogues of $\partial_t P$, $\partial_a P$, $\partial_b P$ together with the mixed derivatives $\partial_t \partial_a P$, $\partial_t \partial_b P$, $\partial_a \partial_b P$. Particular attention is given to $\partial_t \partial_b P$, which measures how the temporal evolution of an instrument differs across broker observations.

The second part of the framework introduces a quasiconformal geometric layer. Local market patches are normalized and compared by deformation maps; the Beltrami coefficient $\mu = f_{\bar{z}}/f_z$ and the quasiconformal dilatation $K = (1 + |\mu|)/(1 - |\mu|)$ provide candidate measures of structural deformation. The proposed financial interpretation is deliberately conservative: the market is not assumed to literally be a Riemann surface. Instead, market data are embedded into finite-dimensional geometric patches, and quasiconformal mappings are used as a mathematical comparison device.

The literature already contains geometric approaches to finance — manifold learning, Riemannian and Minkowski formulations, topological and cohomological constructions, and a direct use of Teichmüller and moduli-space ideas in financial time-series classification. The proposed contribution is therefore not "geometry in finance" in general; the specific gap addressed is the *combination* of synchronized multi-asset observations, multiple observation venues, explicit quasiconformal deformation, and a Teichmüller-type market distance.

The resulting framework defines a market state in terms of geometric deformation, cross-asset coherence, cross-broker coherence, mixed-dimensional coupling, and triangular consistency, and formulates empirical hypotheses, null models, estimator validation, falsification tests, and potential applications to regime detection. The primary research question is not whether a new indicator can be invented, but whether the multidimensional geometry of synchronized market observations contains information that cannot be reduced to volatility, correlation, PCA, cointegration, or standard technical indicators.

Contents

1 Introduction

Conventional financial chart analysis begins with a one-dimensional time series $P(t)$. The independent variable is time and the dependent variable is price. From this representation one constructs moving averages, oscillators, volatility estimators, channels, trend lines, and other derived statistics.

This representation is useful, but it removes information that is intrinsic to real markets. Foreign exchange makes the loss especially visible and serves as the running example; the construction is not restricted to it (Section 51).

A currency pair is not an isolated object. It is related to other currency pairs through currency identities, triangular relationships, liquidity conditions, funding relationships, order flow, and market microstructure. A metal, an index, or a crypto-asset is likewise coupled to the rest of the universe — through common risk factors, funding currencies, and shared liquidity — even where no exact algebraic identity exists.

Furthermore, the same nominal instrument can be observed through multiple brokers or liquidity sources. These observations are similar but generally not identical.

Consequently, instead of considering only $P(t)$, we propose the multidimensional object

$$\boxed{P(t, a, b)}$$

with

$$t: \text{ time}, \qquad a: \text{ asset / instrument}, \qquad b: \text{ broker / observation source}.$$

This creates three distinct directions. For fixed (a, b), the map $t \mapsto P(t, a, b)$ is the conventional price chart. For fixed (t, a), the map $b \mapsto P(t, a, b)$ describes the cross-broker realization of the same instrument at the same moment. For fixed (t, b), the map $a \mapsto P(t, a, b)$ describes the cross-asset market state.

The complete object therefore contains substantially more structure than an ordinary chart. The main proposition of this paper is:

> The geometry of the multidimensional observation structure may contain information about market regimes that is not visible in an individual price path.

The proposed framework is deliberately structure-first. No trading rule is assumed in advance. The research sequence is

market data $\rightarrow$ geometric representation $\rightarrow$ deformation $\rightarrow$ market state $\rightarrow$ empirical validation.

2 Related Work and the Research Gap

Geometric approaches to finance are established, and the present framework must be positioned against them rather than beside them. Differential-geometric methods have long been used in asset pricing and incomplete-market models, while more recent work embeds financial observations into low-dimensional manifolds. A directly relevant recent example is the use of constant-curvature two-manifolds as latent geometries for financial data, with spherical, Euclidean, hyperbolic, and toroidal candidates compared empirically [15]. Pinčák and Kanjamapornkul explicitly couple price and time through a Minkowski metric in a GARCH framework [16], and a cohomological and topological treatment of financial time series — including link and knot structures and a topological interpretation of bid–ask dynamics — has also been proposed [13].

The closest direct prior art identified is the *Support Spinor Machine* [12], which uses moduli-space and Teichmüller-space ideas as part of a nonlinear time-series classification architecture.

It would therefore be incorrect to claim that Teichmüller concepts have never been applied to financial time series. The claim of this paper is narrower. The intended research gap is

$$\boxed{\begin{array}{c}\text{synchronized multi-asset observations} \times \text{multiple venues} \\ \times \text{ explicit quasiconformal deformation} \times \text{Teichmüller-type market distance}\end{array}}$$

— that is, the specific combination, not any individual ingredient.

Classical quasiconformal and Teichmüller theory provides the mathematical foundation; the standard modern framework is Gardiner and Lakic [11], with [1] and [2] as classical references. In local complex coordinates, a quasiconformal map satisfies

$$f_{\bar{z}} = \mu(z)\, f_z, \qquad \|\mu\|_\infty < 1, \tag{2.1}$$

with the local distortion $K(z)$ developed in Section 16.

To keep the boundary between foundation and proposal visible, every construction in this paper is assigned one of three statuses — *established mathematics*, *proposed construction*, or *research hypothesis* — and Appendix E records this assignment per component.

3 The Market Tensor $P(t, a, b)$

Let $\mathcal{T} = \{t_1, \dots, t_T\}$ be a common time grid, let $\mathcal{A} = \{a_1, \dots, a_N\}$ be the set of instruments — in the FX example currency pairs, in general any mix of asset classes — and let $\mathcal{B} = \{b_1, \dots, b_M\}$ be the set of brokers or independent price-feed sources.

The synchronized price tensor is defined as

$$P_{ijk} = P(t_i, a_j, b_k). \tag{3.1}$$

Thus the complete data structure has $T \times N \times M$ price observations before considering bid, ask, OHLC, volume, spread, or order-book information.

Definition 3.1 (Market Tensor)**.** *The synchronized market tensor is*

$$\mathcal{X} = \{P(t, a, b) \mid t \in \mathcal{T},\ a \in \mathcal{A},\ b \in \mathcal{B}\}.$$

The tensor should be interpreted as an observation structure rather than as a conventional three-dimensional physical space. The coordinates have different meanings; (t, a, b) is therefore *not* symmetric under arbitrary coordinate permutations.

4 Temporal Synchronization of Multi-Venue Data

Cross-broker comparison requires a common temporal coordinate. Suppose broker b_1 constructs a five-minute bar over $[12{:}00, 12{:}05)$ while broker b_2 constructs a bar over $[12{:}01, 12{:}06)$. The resulting bars cannot be compared directly without introducing an artificial phase displacement.

We therefore define a canonical partition $\mathcal{I} = \{I_1, \dots, I_T\}$ and a synchronization operator

$$\mathscr{S} : P^{(b)} \mapsto P^{(b)}_{\mathcal{I}}.$$

All brokers must be mapped onto the same canonical intervals.

For tick data, the synchronization problem is more subtle because quotes arrive asynchronously. Possible approaches include:

1. previous-tick interpolation;
2. linear interpolation;

3. event-time sampling;
4. nearest-neighbor synchronization;
5. volume-clock or transaction-clock synchronization.

The chosen procedure must be fixed before statistical testing.

5 Logarithmic Price Coordinates

Prices are multiplicative quantities. Define

$$x(t, a, b) = \log P(t, a, b). \tag{5.1}$$

The temporal return becomes

$$r_t(t, a, b) = x(t + \Delta t, a, b) - x(t, a, b). \tag{5.2}$$

Logarithmic coordinates are particularly useful because cross-rate relationships become additive. For example, EUR/JPY = (EUR/USD)(USD/JPY) implies

$$\log(\text{EUR/JPY}) = \log(\text{EUR/USD}) + \log(\text{USD/JPY}).$$

This provides a natural connection between the asset dimension and the underlying currency dimension.

6 Currency-Network Representation

This section and the two that follow exploit relational structure that is specific to currencies. In a broader universe they apply to the FX subgraph; instruments without exact algebraic cross-identities simply contribute no closure constraints, while the temporal and broker dimensions apply to them unchanged.

Let $\mathcal{C} = \{c_1, \dots, c_K\}$ denote the currency universe. An FX pair i/j can be represented through latent currency coordinates $u_i(t, b)$ and $u_j(t, b)$. Define

$$x_{ij}(t, b) = u_i(t, b) - u_j(t, b) + \varepsilon_{ij}(t, b). \tag{6.1}$$

The residual ε_{ij} contains quotation noise, microstructure effects, timing effects, and deviations from the latent currency representation.

The representation has a gauge freedom because $(u_i + \lambda) - (u_j + \lambda) = u_i - u_j$. We therefore impose

$$\sum_{i=1}^{K} u_i(t, b) = 0. \tag{6.2}$$

This converts the currency system into a normalized coordinate system.

7 The Currency Graph and Its Incidence Matrix

Let $G_A = (V_A, E_A)$ be the directed currency graph. Currencies form vertices, $V_A = \mathcal{C}$; FX pairs form directed edges, $E_A \subseteq \mathcal{C} \times \mathcal{C}$.

Let B denote the incidence matrix. Then the vector of pairwise log prices may be approximated by

$$\boldsymbol{x}(t, b) = B\boldsymbol{u}(t, b) + \boldsymbol{\varepsilon}(t, b). \tag{7.1}$$

This representation is useful because it allows the cross-asset layer to be interpreted as a projection of a smaller currency state.

8 Triangular No-Arbitrage Consistency

For three currencies i, j, k, the identity $x_{ij} + x_{jk} + x_{ki} = 0$ holds in the idealized frictionless case. Define the triangular residual

$$\tau_{ijk}(t, b) = x_{ij}(t, b) + x_{jk}(t, b) + x_{ki}(t, b). \tag{8.1}$$

In real markets $\tau_{ijk} \neq 0$ because of spreads, transaction costs, latency, quotation conventions, and liquidity constraints. The residual should therefore be treated as a structural observable, not as an automatic arbitrage signal.

Exact closure identities of this kind are a privilege of the currency subgraph. Other asset classes contribute analogous but weaker consistency relations — the spot–futures basis of a metal or index, or the cross-venue identity of the same crypto-asset — and where no such relation exists, the venue dimension alone supplies the consistency structure: the same instrument observed twice must agree up to the broker operators of Section 35.

9 The Three Geometric Directions: Time, Asset, Broker

The market tensor possesses three fundamental directional variations.

9.1 Temporal direction

The discrete temporal derivative is

$$\Delta_t P_{ijk} = P_{i+1,j,k} - P_{i,j,k}. \tag{9.1}$$

This is the ordinary direction of chart movement.

9.2 Asset direction

For appropriately normalized asset coordinates,

$$\Delta_a P_{ijk} = P_{i,j+1,k} - P_{i,j,k}. \tag{9.2}$$

This represents cross-asset structural variation. Because arbitrary numerical ordering of assets is generally meaningless, a practical implementation should preferably replace the raw index j with a graph, factor, or manifold coordinate.

9.3 Broker direction

For fixed time and asset,

$$\Delta_b P_{ijk} = P_{i,j,k+1} - P_{i,j,k}. \tag{9.3}$$

This is the crucial observation dimension. It is not another time series. It represents the dispersion of nominally identical market observations.

10 The Broker Fiber: Cross-Venue Price Dispersion

For fixed (t, a) define the broker fiber

$$\mathcal{F}_{t,a} = \{P(t, a, b) : b \in \mathcal{B}\}. \tag{10.1}$$

A robust central observation is

$$\bar{x}(t, a) = \mathrm{median}_b\ x(t, a, b). \tag{10.2}$$

The broker deviation is

$$d_b(t,a,b) = x(t,a,b) - \bar{x}(t,a). \tag{10.3}$$

A robust dispersion measure is

$$\sigma_b(t,a) = 1.4826\ \mathrm{median}_b \left| d_b(t,a,b) \right|. \tag{10.4}$$

The coefficient 1.4826 converts the MAD to the standard-deviation scale under Gaussian assumptions.

11 Mixed Temporal–Broker Geometry

The most important broker statistic is not simply $\Delta_b P$. Instead, consider its temporal evolution. Define

$$\Delta_t \Delta_b P_{ijk} = \left(P_{i+1,j,k+1} - P_{i+1,j,k}\right) - \left(P_{i,j,k+1} - P_{i,j,k}\right). \tag{11.1}$$

In continuous notation, $\partial_t \partial_b P$. This quantity answers:

> Do two broker observations evolve differently during the same market movement?

A small broker spread that remains structurally stable may be uninteresting. A rapidly changing broker differential may indicate:

- liquidity fragmentation;
- feed latency;
- different liquidity providers;
- spread expansion;
- asynchronous quote updates;
- market microstructure stress;
- timestamp problems.

12 Cross-Asset Temporal Geometry

Analogously, $\Delta_t \Delta_a P$ measures how temporal movement differs across assets. This is fundamentally different from simple correlation.

Correlation asks: *do two return series move together?* The mixed derivative asks: *how does the local temporal geometry vary across the asset dimension?*

The distinction becomes important during transitions. Two assets may retain high correlation while their local geometric structure changes dramatically.

13 First- and Second-Order Differential Structure

The differential of the market field is formally

$$dP = P_t\,dt + P_a\,da + P_b\,db. \tag{13.1}$$

The second-order differential contains

$$\begin{aligned} d^2P = {} & P_{tt}\,dt^2 + P_{aa}\,da^2 + P_{bb}\,db^2 \\ & + 2P_{ta}\,dt\,da + 2P_{tb}\,dt\,db + 2P_{ab}\,da\,db. \end{aligned} \tag{13.2}$$

The six second-order components have different interpretations. P_{tt} describes temporal curvature; P_{aa} describes curvature across assets; P_{bb} describes curvature across broker observations. The mixed components P_{ta}, P_{tb}, P_{ab} describe interactions between the three market dimensions.

14 Local Market Surfaces and the Induced Metric

Fix an asset a and define a broker-time surface

$$S_a(t,b) = x(t,a,b). \tag{14.1}$$

The local tangent vectors are $S_t = \partial_t S$ and $S_b = \partial_b S$. The induced metric is

$$g = \begin{pmatrix} \langle S_t, S_t\rangle & \langle S_t, S_b\rangle \\ \langle S_b, S_t\rangle & \langle S_b, S_b\rangle \end{pmatrix}. \tag{14.2}$$

Define

$$\rho_{tb} = \frac{g_{tb}}{\sqrt{g_{tt}g_{bb}}}. \tag{14.3}$$

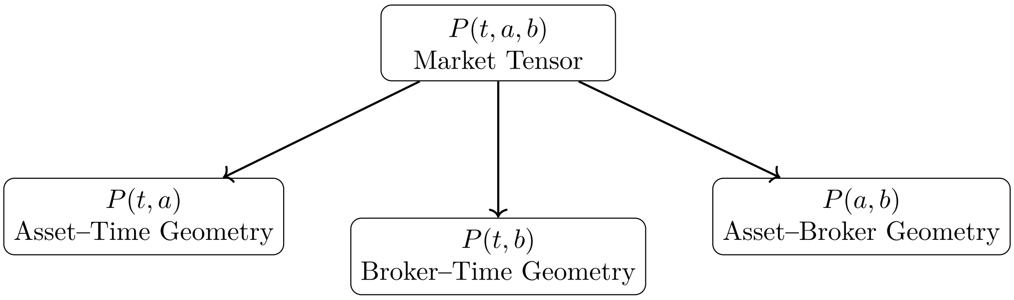
This is a normalized local coupling between temporal and broker variation. Similarly, for fixed broker b, the surface $S_b(t,a) = x(t,a,b)$ produces an asset-time surface, and

$$\rho_{ta} = \frac{g_{ta}}{\sqrt{g_{tt}g_{aa}}} \tag{14.4}$$

measures temporal-asset coupling.

15 The Tensor as a Family of Coupled Surfaces

The three surfaces (t,a), (t,b), and (a,b) are projections of the same underlying tensor.



The complete market geometry should therefore be regarded as a family of coupled surfaces rather than as a single chart.

16 Quasiconformal Maps and the Beltrami Coefficient

A central question is how one local market geometry changes into another. Let Φ_0 be a reference market patch and Φ_1 a later market patch. We seek a map

$$f : \Phi_0 \to \Phi_1. \tag{16.1}$$

For a complex coordinate $z = x + iy$, a sufficiently regular map can be written locally as $df = f_z\, dz + f_{\bar z}\, d\bar z$. The Beltrami coefficient is

$$\mu(z) = \frac{f_{\bar z}(z)}{f_z(z)}. \tag{16.2}$$

For an orientation-preserving quasiconformal map, $|\mu(z)| < 1$. The local dilatation is

$$K(z) = \frac{1 + |\mu(z)|}{1 - |\mu(z)|}. \tag{16.3}$$

When $\mu = 0$ the transformation is locally conformal and $K = 1$. As $|\mu| \to 1$, the distortion increases.

17 Financial Interpretation of the Beltrami Field

The proposed interpretation is

$$\boxed{\mu \quad \leftrightarrow \quad \text{local market deformation}}$$

and

$$\boxed{K \quad \leftrightarrow \quad \text{magnitude of structural distortion.}}$$

This interpretation must be treated as a modeling construction. It is *not* claimed that the observed market is intrinsically a complex manifold. Instead, the financial data are embedded into a local two-dimensional coordinate patch.

One possible coordinate system is $z = u + iv$ with u a normalized time and v a normalized cross-sectional coordinate. The cross-sectional coordinate may represent:

1. broker rank;
2. a latent currency factor;
3. a PCA coordinate;
4. a diffusion-map coordinate;
5. a graph coordinate.

18 A Teichmüller-Type Deformation Distance

Let Φ_0, Φ_1 be normalized market patches. Define the minimal quasiconformal distortion

$$K_{\min} = \inf_f \sup_z K_f(z), \tag{18.1}$$

where f ranges over the admissible deformation maps. We then define the proposed geometric distance

$$D_T(\Phi_0, \Phi_1) = \frac{1}{2} \log K_{\min}. \tag{18.2}$$

This is inspired by the standard logarithmic quasiconformal metric structure of Teichmüller theory. The financial quantity should therefore be interpreted as a *Teichmüller-inspired deformation statistic*.

19 Market Teichmüller Space

Let $\mathfrak{M}$ denote the set of normalized market patches. Introduce an equivalence relation $\Phi_1 \sim \Phi_2$ when the two patches differ only through the chosen admissible normalizations and conformal transformations. The resulting quotient is denoted

$$\mathcal{T}_{\mathrm{mkt}} = \mathfrak{M}/\sim. \tag{19.1}$$

A market state becomes $[\Phi_t] \in \mathcal{T}_{\mathrm{mkt}}$, and a market trajectory is $t \mapsto [\Phi_t]$. The corresponding geometric velocity is

$$V_G(t) = \frac{D_T(\Phi_t, \Phi_{t+\Delta t})}{\Delta t}. \tag{19.2}$$

The geometric acceleration is

$$A_G(t) = V_G(t) - V_G(t - \Delta t). \tag{19.3}$$

20 Price Movement versus Structural Movement

A geometrically stable market regime satisfies approximately $D_T(\Phi_t, \Phi_{t+\Delta t}) \approx 0$. A transition satisfies $D_T(\Phi_t, \Phi_{t+\Delta t}) \uparrow$. A rapidly developing structural transition satisfies $A_G(t) \gg 0$.

This gives a conceptual distinction between price movement and structural movement. A market may exhibit $|\Delta P| \gg 0$ while $D_T \approx 0$. This would correspond to a strong movement that preserves the local market geometry.

Conversely, $|\Delta P| \approx 0$ may coexist with $D_T \gg 0$. This would indicate a structural transition occurring before a large-amplitude price movement.

21 Cross-Broker Coherence

Define broker return dispersion

$$\sigma_B^2(t,a) = \mathrm{Var}_b\left[r_t(t,a,b)\right]. \tag{21.1}$$

A normalized coherence statistic may be defined as

$$C_B(t,a) = 1 - \frac{\sigma_B(t,a)}{\sigma_{\mathrm{ref}}(t,a)+\epsilon}. \tag{21.2}$$

The precise bounded normalization should be chosen empirically. A high value indicates that the temporal realization of the same instrument is similar across brokers; a low value indicates divergence.

22 Cross-Asset Coherence

Let $\boldsymbol{r}_t$ be the vector of normalized asset returns and let $\Sigma_t = \mathrm{Cov}(\boldsymbol{r}_t)$ be the local covariance matrix. The first principal component explains

$$C_A(t) = \frac{\lambda_1(\Sigma_t)}{\mathrm{tr}(\Sigma_t)}. \tag{22.1}$$

This is a benchmark measure of cross-asset coherence. The proposed geometric framework should be tested *against* this quantity rather than simply replacing it.

23 Joint Market Coherence

A simple composite statistic is

$$C_{AB}(t) = C_A(t)C_B(t). \tag{23.1}$$

A more rigorous implementation would estimate a joint latent coherence factor. The conceptual interpretation is that $C_{AB} \approx 1$ means the market is coherent across both assets and broker observations.

24 The Geometric Market State Vector

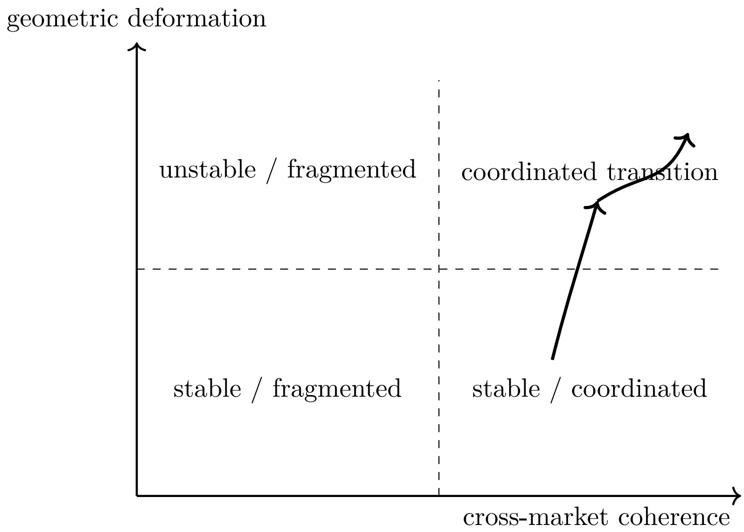
Define the state vector

$$\mathbf{S}_t = \begin{pmatrix} D_T(t) & V_G(t) & A_G(t) & C_A(t) & C_B(t) & D_{ta}(t) & D_{tb}(t) & \tau(t) \end{pmatrix}^\top. \tag{24.1}$$

This vector is the central candidate representation of market state. It deliberately avoids reducing the market to a single scalar.

25 The Coherence–Deformation Phase Diagram

A particularly useful projection is $x = C_{AB}$ against $y = D_T$. This produces the qualitative phase structure below.



The upper-right region is especially important. It represents $C_{AB} \gg 0$ combined with $D_T \gg 0$ — many observations changing coherently at the same time. Such a state may represent a broad regime transition.

26 A Geometry-Driven Dynamic Price Channel

The framework can be used to construct a dynamic channel. Let m_t be a structural centerline. Define the geometric width

$$w_t = w_0 \exp\left(\alpha D_T(t) + \beta V_G(t) + \gamma \sigma_t\right). \tag{26.1}$$

The channel boundaries are $U_t = m_t + w_t$ and $L_t = m_t - w_t$. A more general asymmetric construction is

$$U_t = m_t + w_t^+, \qquad L_t = m_t - w_t^-. \tag{26.2}$$

The widths may depend separately on the direction of deformation.

27 Beltrami Phase and Directional Asymmetry

The magnitude $|\mu|$ describes the amount of deformation. The phase $\arg(\mu)$ contains directional information in the complex representation. This suggests an extension

$$w_t^\pm = f^\pm\left(|\mu_t|, \arg(\mu_t), C_t\right). \tag{27.1}$$

The hypothesis is that the *direction* of geometric deformation may contain information that is discarded when only a scalar volatility measure is used. This must be empirically tested.

28 The Three Deformation Families

The framework distinguishes three primary deformations.

28.1 Temporal deformation

$$D_t = D_T(\Phi_t, \Phi_{t+\Delta t}).$$

28.2 Asset deformation

For assets a_i, a_j,

$$D_a(a_i, a_j; t) = D_T(\Phi_{a_i,t}, \Phi_{a_j,t}). \tag{28.1}$$

28.3 Broker deformation

For brokers b_i, b_j,

$$D_b(b_i, b_j; a, t) = D_T(\Phi_{a,b_i,t}, \Phi_{a,b_j,t}). \tag{28.2}$$

These quantities should not be conflated.

29 Mixed Deformation Statistics

The most informative statistics may be mixed. Define D_{ta}, D_{tb}, D_{ab}, whose discrete counterparts are based on $\Delta_t\Delta_a P$, $\Delta_t\Delta_b P$, and $\Delta_a\Delta_b P$.

The temporal-broker term D_{tb} is especially important because it distinguishes common market movement from observation-specific movement. The temporal-asset term D_{ta} distinguishes common factor movement from pair-specific movement. The asset-broker term D_{ab} tests whether apparent asset relationships are stable across observation sources.

30 A Discrete Quasiconformal Approximation via the Jacobian

A full complex quasiconformal model is computationally demanding. A practical first implementation can use the local Jacobian

$$J = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}.$$

Let its singular values be $s_1 \geq s_2 > 0$. Define

$$K_{\text{disc}} = \frac{s_1}{s_2}. \tag{30.1}$$

The corresponding deformation is

$$D_{\text{disc}} = \frac{1}{2}\log K_{\text{disc}}. \tag{30.2}$$

This gives a real-valued discrete analogue of quasiconformal dilatation. Only after establishing empirical value should the full Beltrami optimization be introduced.

31 Estimating a Financial Beltrami Field

Let $z = u + iv$ be a local market coordinate, $F_0(u, v)$ a reference market patch and $F_1(u, v)$ the observed patch. We seek $f : F_0 \to F_1$. The map may be estimated through

$$\hat{f} = \arg\min_f \left[\mathcal{L}_{\text{data}}(f) + \lambda\, \mathcal{R}_{\text{QC}}(f)\right]. \tag{31.1}$$

Here $\mathcal{L}_{\text{data}}$ measures reconstruction error and $\mathcal{R}_{\text{QC}}$ regularizes the deformation. The resulting

$$\hat{\mu} = \frac{\partial_{\bar{z}}\hat{f}}{\partial_z \hat{f}} \tag{31.2}$$

defines the empirical Beltrami field.

Admissibility and guard discipline

A market patch is admissible for this estimation only if explicit data-quality conditions hold:

1. synchronized timestamps;
2. stable instrument identity;
3. stable venue identity;
4. cross-rate consistency checks;
5. explicit missing-data treatment;
6. explicit price definition (bid, ask, mid, last, mark);
7. numerical validity of the quasiconformal estimate.

The last condition is a hard guard. If the estimated field violates $\|\hat{\mu}_t\|_\infty < 1$ on some region, no standard quasiconformal interpretation may be claimed on that domain — the estimator must flag and exclude it, not clip the value back into range. An admissibility failure is a result about the data, not a numerical nuisance to be smoothed away.

32 Surface Geometry and the Metric Tensor

For a broker-time surface $S(t, b)$, the induced metric is

$$g_{\alpha\beta} = \langle \partial_\alpha S, \partial_\beta S \rangle, \qquad \alpha, \beta \in \{t, b\}. \tag{32.1}$$

The determinant $\det g = g_{tt}g_{bb} - g_{tb}^2$ measures local area scaling; the off-diagonal g_{tb} measures directional coupling; and the normalized form is $\rho_{tb} = g_{tb}/\sqrt{g_{tt}g_{bb}}$. The same construction can be applied to the (t, a) and (a, b) surfaces.

33 Market Geometry as a Fibered Structure

The tensor $P(t, a, b)$ can be interpreted as a family of broker fibers over the asset-time base:

$$\pi : (t, a, b) \mapsto (t, a). \tag{33.1}$$

For every (t, a) there exists a fiber $\mathcal{F}_{t,a}$. This suggests a fibered market geometry:

$$\boxed{\text{base} = \text{time–asset structure}} \qquad \boxed{\text{fiber} = \text{broker realization.}}$$

The geometry of the fibers measures observation dispersion, while the geometry of the base measures market structure. The mixed derivative $\partial_t\partial_b P$ then measures how the fiber geometry evolves as the base moves through time.

34 Recovering a Latent Common Market Geometry

Suppose several brokers observe the same latent market process $X(t, a)$. Each broker produces

$$P(t, a, b) = \mathcal{O}_b[X(t, a)] + \eta_b(t, a), \tag{34.1}$$

where $\mathcal{O}_b$ is the broker observation operator and η_b is observation noise. The research problem becomes:

> Can the common latent geometry X be recovered from multiple deformed observations?

This is a central conceptual bridge between multi-broker analysis and deformation geometry.

35 Broker Observation Operators

A broker may be modeled abstractly as

$$\mathcal{O}_b = \mathcal{A}_b \circ \mathcal{L}_b \circ \mathcal{T}_b, \tag{35.1}$$

where $\mathcal{T}_b$ is the timestamp transformation, $\mathcal{L}_b$ the liquidity and aggregation transformation, and $\mathcal{A}_b$ the quote construction. The observed price becomes $P_b = \mathcal{O}_b[X]$.

The objective is not necessarily to recover each operator exactly. Instead, the geometric framework attempts to identify the portion of the observed structure that is *invariant* across operators.

36 Cross-Broker Consensus Geometry

Let $\Phi_{b_1}, \dots, \Phi_{b_M}$ be broker-specific market patches. Define a consensus patch

$$\Phi^\star = \arg\min_{\Phi} \sum_{b=1}^{M} D_T(\Phi, \Phi_b)^2. \tag{36.1}$$

This is analogous to a geometric Fréchet mean. Each broker then has a deformation distance $D_b = D_T(\Phi_b, \Phi^\star)$. The consensus geometry provides a reference against which broker-specific deviations can be measured.

A linear alternative: the market compensator

For two admissible observation representations $\Phi^{(1)}, \Phi^{(2)}$ — for example two venue realizations of the same instrument — let J denote the involution exchanging them. Define the compensator

$$\mathsf{C}_{\text{mkt}} = \frac{1}{2}(\mathsf{I} + \mathsf{J}), \tag{36.2}$$

so that $\mathsf{C}_{\text{mkt}}(\Phi) = \frac{1}{2}(\Phi^{(1)} + \Phi^{(2)})$, and the antisymmetric residual

$$\mathsf{R}_{\text{mkt}} = \frac{1}{2}(\mathsf{I} - \mathsf{J})\Phi, \qquad \Phi = \mathsf{C}_{\text{mkt}}\Phi + \mathsf{R}_{\text{mkt}}. \tag{36.3}$$

The symmetric component is a candidate common-market estimate — a linear counterpart of the Fréchet-type consensus patch (36.1) that requires no optimization. The residual records observation-specific structure.

Critically, $\mathsf{R}_{\text{mkt}} = 0$ must never be inferred merely because the compensator has been *defined*. It must be measured, or proved under explicit assumptions.

37 Cross-Asset Consensus Geometry

The same construction can be applied across assets. Given $\Phi_{a_1}, \dots, \Phi_{a_N}$, define

$$\Phi_A^\star = \arg\min_{\Phi} \sum_{a=1}^{N} D_T(\Phi, \Phi_a)^2. \tag{37.1}$$

The residual $D_a = D_T(\Phi_a, \Phi_A^\star)$ measures how far an individual asset deviates from the common market geometry. This can be particularly useful for identifying relative strength and relative structural isolation.

38 Coordinated and Fragmented Regimes

The combination of coherence and deformation leads to four qualitative states.

Coherence	Deformation	Interpretation
High	Low	Stable coordinated regime
High	High	Coordinated transition
Low	Low	Fragmented / range regime
Low	High	Structural instability

The final state is particularly important. If $D_T \uparrow$ while $C_{AB} \downarrow$, then the market is becoming geometrically unstable without developing a coherent cross-market direction. This may correspond to noise, liquidity stress, or the early phase of a regime transition.

39 Research Hypotheses

Hypothesis 1 (Cross-broker coherence). *After correct temporal synchronization and normalization, liquid instruments exhibit significant cross-broker structural coherence relative to randomized broker assignments.*

Hypothesis 2 (Geometric transition). *The deformation statistic D_T increases systematically around independently identified market regime transitions.*

Hypothesis 3 (Incremental information). *The geometric state vector contains information not explained by standard volatility, correlation, PCA, cointegration, or technical indicators.*

Hypothesis 4 (Coordinated transition). *Periods with simultaneously high coherence and increasing geometric deformation are disproportionately associated with broad market transitions.*

Hypothesis 5 (Broker anomaly detection). *Broker-specific geometric deviations can identify feed-specific anomalies more effectively than simple absolute price differences.*

Hypothesis 6 (Triangular closure). *Changes in the geometry of the FX triangular residual τ precede or coincide with selected liquidity and volatility transitions.*

Hypothesis 7 (Coordinate robustness). *A properly defined geometric deformation score is more stable under admissible coordinate changes than raw-price indicators.*

40 Comparison with Correlation, PCA, Cointegration, and Optimal Transport

40.1 Correlation

Correlation measures linear dependence,

$$\rho_{ij} = \frac{\mathrm{Cov}(r_i, r_j)}{\sigma_i \sigma_j}.$$

It does not directly measure local geometric deformation.

40.2 Principal component analysis

PCA solves $\Sigma v_k = \lambda_k v_k$. The proposed framework may use PCA coordinates but introduces additional deformation information. PCA should therefore be a benchmark, not ignored.

40.3 Cointegration

Cointegration identifies stable long-run combinations. The proposed framework can incorporate cointegration residuals as coordinates but does not reduce the market geometry to them.

40.4 Optimal transport

Optimal transport provides a powerful alternative way of comparing distributions of market states. It should be included as a benchmark in any serious empirical test.

40.5 Manifold learning

Diffusion maps, Isomap, UMAP, and related methods can recover nonlinear low-dimensional structure. The Teichmüller component is justified only if it provides incremental information or a theoretically useful deformation metric.

41 Null Models and Estimator Validation

A sophisticated geometric model is vulnerable to false discoveries. The following null models are therefore mandatory, together with two positive validation tests of the estimator itself.

41.1 Broker permutation

Randomly permute broker identities while preserving the temporal and asset marginal distributions. If the geometric signal remains unchanged, the broker dimension contains no meaningful information.

41.2 Time shuffle

Randomly permute temporal ordering within blocks. A temporal geometric signal should be destroyed.

41.3 Asset shuffle

Randomly permute asset identities while preserving marginal return distributions. This tests whether cross-asset geometry depends on actual market relationships.

41.4 Synthetic models

At minimum test:

1. independent random walks;
2. correlated Gaussian processes;
3. stochastic volatility;
4. jump-diffusion;
5. regime-switching processes;
6. latent-factor models with broker-specific observation noise.

41.5 Synthetic deformation recovery

Before any market data are touched, generate a known quasiconformal map with a prescribed Beltrami field and verify that the estimation pipeline of Section 31 recovers it. An estimator that cannot recover a known μ from clean synthetic data has no business interpreting noisy market data. This test is logically prior to every null model above: the null models ask whether the market contains structure; this test establishes whether the instrument can measure structure at all.

41.6 Coordinate-invariance test

The embedding is determined only up to admissible conformal transformations (Sections 16 and 19). Apply such transformations to the same data and verify that the intrinsic deformation score is unchanged. A score that varies under conformal coordinate changes measures the choice of embedding, not the market — and every downstream result inherits that defect.

42 Empirical Dataset: FX Core and Cross-Asset Extension

A first empirical experiment should contain at least $M \geq 3$ independent broker or price-feed sources and $N \geq 10$ liquid instruments. A core FX universe may include

$$\begin{array}{lllll} \text{EUR/USD}, & \text{GBP/USD}, & \text{USD/JPY}, & \text{USD/CHF}, & \text{AUD/USD}, \\ \text{USD/CAD}, & \text{NZD/USD}, & \text{EUR/JPY}, & \text{EUR/GBP}, & \text{GBP/JPY}. \end{array}$$

This universe should be extended by liquid instruments from outside FX that are observed through the same venues — for example XAU/USD, a major equity-index CFD, and BTC/USD — so that the asset coordinate spans several asset classes and cross-asset findings are demonstrably not an artifact of currency structure alone. The currency-network machinery of Sections 6–8 then applies to the FX subgraph, while every instrument participates in the temporal and broker dimensions.

Five-minute bars provide an accessible initial test. Tick-level data should be used for a second-stage microstructure analysis.

43 Rolling Window Estimation

For every time t, define $W_t = [t - L, t]$. All geometric quantities must be estimated using information in W_t only. This includes:

- normalization parameters;
- reference geometries;
- PCA coordinates;
- deformation parameters;
- latent factors;
- model hyperparameters.

The resulting state $\mathbf{S}_t$ must therefore be measurable with respect to information available at time t.

44 Data Leakage and Look-Ahead Bias

The framework is especially vulnerable to look-ahead bias because geometric normalization may inadvertently use future observations. For a valid backtest,

$$\theta_t = F(P_s : s \leq t) \tag{44.1}$$

must hold for every estimated parameter θ_t. The future return $r_{t+\Delta}$ may only be used for evaluation. No future data may influence Φ_t, μ_t, K_t, C_t, or D_t.

45 Trading Interpretation: From Geometry to Regime

The proposed geometry should *not* directly predict price direction. A safer hierarchy is

$$\boxed{\text{geometry} \to \text{regime} \to \text{conditional strategy} \to \text{risk management}}$$

For example, $D_T \downarrow$ together with $C_{AB} \uparrow$ could indicate a stable coordinated regime. A transition state could satisfy $D_T \uparrow$, $C_{AB} \uparrow$. A fragmented state could satisfy $D_T \uparrow$, $C_{AB} \downarrow$.

Direction would then be supplied by a separate directional model. This prevents the geometric framework from being forced to answer a question it was not designed to answer.

46 Transaction Costs and Broker Reality

Broker-level differences must be interpreted carefully. A deviation ΔP is not economically exploitable unless

$$|\Delta P| > \text{spread} + \text{commission} + \text{slippage} + \text{latency costs} + \text{other execution costs}.$$

Therefore:

$$\boxed{\text{geometric broker divergence} \neq \text{arbitrage opportunity}}$$

The primary purpose of the broker dimension is structural measurement.

47 Feed Anomaly Detection

A useful secondary application is detection of feed-specific anomalies. Suppose B_1, B_2, B_3 observe approximately the same market movement but B_1 deviates strongly. Define

$$A_b(t, a) = D_T(\Phi_{a,b,t}, \Phi^\star_{a,t}). \tag{47.1}$$

A large A_b conditional on high cross-broker consensus may indicate a broker-specific anomaly. This can be used as a data-quality layer before any trading model.

48 Market Regime Classification

The geometric state vector can be used in a classifier $R_t = f(\mathbf{S}_t)$. Possible regimes include

$$R_t \in \{\text{trend}, \text{range}, \text{transition}, \text{fragmentation}, \text{event}, \text{microstructure stress}\}.$$

The classifier should preferably be trained using independent labels — volatility regimes, structural breaks, or event windows — rather than defining labels from the same geometric variables.

49 Multi-Timeframe Extension

Introduce a scale coordinate $\ell \in \{\text{tick}, 1\text{s}, 1\text{m}, 5\text{m}, 1\text{h}, 1\text{d}\}$. The market tensor becomes

$$P(t, a, b, \ell). \tag{49.1}$$

The deformation field is now $D(t, a, b, \ell)$. A particularly interesting question is whether deformation propagates across time scales, $D_{\ell_1} \to D_{\ell_2} \to D_{\ell_3}$. A structural transition that appears simultaneously across several scales may be qualitatively different from a purely local microstructure event.

50 Order-Book and Liquidity-Depth Extension

If depth-of-market information is available, introduce liquidity depth q. The market tensor becomes

$$P(t, a, b, q). \tag{50.1}$$

The q dimension represents the order-book layer. The full structure could then distinguish *price geometry* from *liquidity geometry*. A transition in price geometry without corresponding liquidity deformation may have a different interpretation from a transition accompanied by severe order-book deformation.

51 Generalization Beyond FX

The framework is not inherently restricted to foreign exchange. The same mathematical architecture can be applied to:

- equities across exchanges;
- index futures across venues;
- commodities across trading venues;
- cryptocurrency markets across exchanges;
- options surfaces;
- yield curves;
- volatility surfaces.

The observation coordinate b can represent a broker, an exchange, a venue, or a data vendor. The asset coordinate can represent an instrument, a maturity, or a strike, depending on the market.

52 The Geometric Hierarchy: From Price Chart to Market Tensor

The framework can be summarized by the hierarchy $P(t)$ for an isolated chart, $P(t, a)$ for a multi-asset market, and

$$\boxed{P(t, a, b)}$$

for a multi-asset, multi-observer market. The final object is not merely a collection of charts. It contains directional derivatives P_t, P_a, P_b, mixed derivatives P_{ta}, P_{tb}, P_{ab}, and local metric structure.

The Teichmüller layer adds a deformation comparison $\Phi_t \to \Phi_{t+\Delta t}$. The resulting quantity $D_T = \frac{1}{2}\log K$ is intended to measure structural deformation. The essential distinction is therefore

$$\boxed{\text{price movement} \neq \text{geometric movement}}$$

and

$$\boxed{\text{broker dispersion} \neq \text{temporal movement.}}$$

This distinction is the foundation of the framework.

53 Research Programme

The proposed research programme consists of six stages.

Stage I — Data geometry

Construct $P(t, a, b)$ with rigorous synchronization.

Stage II — Classical geometry

Estimate P_t, P_a, P_b and the mixed terms.

Stage III — Surface geometry

Construct the surfaces (t, a), (t, b), (a, b) and estimate their local metrics.

Stage IV — Quasiconformal geometry

Construct normalized market patches and estimate μ, K, D_T.

Stage V — Statistical validation

Compare against volatility, correlation, PCA, cointegration, and technical indicators.

Stage VI — Economic validation

Evaluate out-of-sample regime classification, conditional returns, and risk-adjusted trading performance.

54 Falsification Criteria

The framework should be considered unsuccessful if:

1. geometric statistics are completely explained by volatility;
2. broker geometry disappears after simple spread adjustment;
3. cross-asset geometry adds no information beyond PCA;
4. deformation has no out-of-sample stability;
5. performance disappears after transaction costs;
6. results depend critically on one broker;
7. results depend critically on one arbitrary asset ordering;

8. the apparent signal disappears under reasonable synchronization perturbations.

The principal failure modes behind these criteria are embedding dependence, timestamp artifacts, liquidity-source changes, hyperparameter overfitting, and the false claim that a numerical Beltrami-like quantity is an exact Teichmüller distance. These are design constraints, not footnotes.

These conditions are important because mathematical complexity can create an illusion of explanatory power.

55 Implementation Architecture

A modular implementation can be organized as

```
data/
    raw/
    synchronized/
    normalized/

geometry/
    tensor.py
    currency_graph.py
    surfaces.py
    derivatives.py
    metrics.py

teichmueller/
    patches.py
    beltrami.py
    distortion.py
    distance.py

statistics/
    coherence.py
    regime.py
    null_models.py

backtest/
    signals.py
    execution.py
    costs.py
    evaluation.py
```

The separation is intentional. The data layer should not contain trading assumptions. The geometry layer should not contain performance optimization. The trading layer should consume validated geometric states.

56 Minimal First Experiment

A first test should remain deliberately small. Use $M = 3$ or more brokers and approximately $N = 10$ liquid instruments, FX pairs plus a small number of non-FX instruments. Construct synchronized five-minute bars. For every rolling window:

1. normalize all prices;

2. calculate log returns;
3. estimate broker dispersion;
4. estimate cross-asset covariance;
5. calculate triangular residuals on the FX subgraph;
6. calculate mixed derivatives;
7. construct local surfaces;
8. calculate discrete distortion;
9. calculate coherence;
10. compare the resulting state against subsequent market behavior.

Only after this basic experiment demonstrates an effect should the full quasiconformal optimization be introduced.

57 Expected Contributions

If validated, the framework could contribute four distinct ideas.

57.1 A multidimensional market representation

$P(t, a, b)$ provides a unified representation of time, asset, and observation dimensions.

57.2 Mixed-dimensional market statistics

The quantities P_{ta}, P_{tb}, P_{ab} provide information that ordinary one-dimensional indicators discard.

57.3 Geometric regime measurement

The deformation D_T provides a candidate measure of structural market change.

57.4 Observation-consensus geometry

Multiple brokers provide multiple realizations of the same market, allowing the construction of a consensus market geometry.

58 Limitations

Several limitations must remain explicit.

1. Brokers are not independent observations in a strict statistical sense.
2. Many of the relevant markets — FX, OTC and CFD instruments, crypto-assets — are decentralized, and there is no single canonical “true” price.
3. The definition of a market patch is not unique.
4. The mapping from financial data to a complex coordinate system is a modeling decision.
5. Teichmüller distance is mathematically well-defined only after the underlying objects and admissible transformations have been specified rigorously.

6. Financial time series are nonstationary.
7. Apparent geometric structure may be a repackaging of standard statistical dependence.

These limitations make empirical falsification essential.

59 Conclusion

This paper proposes a structure-first framework for market analysis based on the multidimensional observation tensor

$$\boxed{P(t, a, b)}$$

where t is time, a the asset, and b the broker. The key conceptual step is to recognize that the three dimensions have different meanings. Movement along t is temporal market evolution; movement along a is cross-asset variation; movement along b is cross-observer variation. The mixed terms

$$\boxed{P_{ta}, \quad P_{tb}, \quad P_{ab}}$$

then describe how these dimensions interact.

The framework subsequently introduces local market patches and quasiconformal deformation. The Beltrami coefficient μ measures local deformation, while $K = (1 + |\mu|)/(1 - |\mu|)$ measures its magnitude. The proposed market deformation distance is

$$\boxed{D_T = \frac{1}{2} \log K_{\min}.}$$

The ultimate market state is represented not by one indicator but by a vector containing deformation, coherence, mixed-dimensional coupling, and consistency:

$$\boxed{\mathbf{S}_t = (D_T, V_G, A_G, C_A, C_B, D_{ta}, D_{tb}, \tau)\,.}$$

The central empirical question is whether this state vector contains information about market regimes that cannot be reduced to standard measures such as volatility, correlation, PCA, cointegration, or conventional technical indicators.

The framework therefore makes a deliberately strong claim about the *research question*, but only a deliberately weak claim about the *result*:

$$\boxed{\text{The geometry is a hypothesis to be tested, not a result to be assumed.}}$$

If the empirical tests validate the construction, the next stage is not merely another indicator. It is the construction of a geometrically defined market state space in which trend, transition, fragmentation, and observation-specific distortion become different regions of the same mathematical object.

A Notation

Symbol	Meaning
$P(t, a, b)$	price of asset a at time t observed by broker b
$x(t, a, b)$	logarithmic price
r_t	temporal log return
t	time coordinate
a	asset / instrument coordinate (currency pair in the FX example)

b	broker / observation coordinate
$\mathcal{T}$	common time grid
$\mathcal{A}$	asset universe
$\mathcal{B}$	broker universe
$\mathcal{C}$	currency universe
B	currency graph incidence matrix
τ	triangular consistency residual
Φ_t	normalized market patch
μ	Beltrami coefficient
K	quasiconformal dilatation
D_T	geometric deformation distance
V_G	geometric velocity
A_G	geometric acceleration
C_A	cross-asset coherence
C_B	cross-broker coherence
D_{ta}	temporal-asset deformation
D_{tb}	temporal-broker deformation
D_{ab}	asset-broker deformation
$\mathsf{C}_{\mathrm{mkt}}$	market compensator
$\mathsf{R}_{\mathrm{mkt}}$	venue/asymmetry residual
$\mathcal{T}_{\mathrm{mkt}}$	proposed market Teichmüller space

B Core Equations

The complete framework can be summarized by the following equations.

Market tensor

$$P = P(t, a, b).$$

Logarithmic coordinate

$$x = \log P.$$

First-order geometry

$$dP = P_t\,dt + P_a\,da + P_b\,db.$$

Second-order geometry

$$d^2P = P_{tt}\,dt^2 + P_{aa}\,da^2 + P_{bb}\,db^2 + 2P_{ta}\,dt\,da + 2P_{tb}\,dt\,db + 2P_{ab}\,da\,db.$$

Beltrami coefficient

$$\mu = \frac{f_{\bar{z}}}{f_z}.$$

Dilatation

$$K = \frac{1 + |\mu|}{1 - |\mu|}.$$

Geometric deformation

$$D_T = \frac{1}{2}\log K_{\min}.$$

Geometric velocity

$$V_G = \frac{D_T(\Phi_t, \Phi_{t+\Delta t})}{\Delta t}.$$

Cross-asset coherence

$$C_A = \frac{\lambda_1(\Sigma)}{\mathrm{tr}(\Sigma)}.$$

Broker dispersion

$$\sigma_B^2 = \mathrm{Var}_b(r_t).$$

Market state

$$\mathbf{S}_t = (D_T, V_G, A_G, C_A, C_B, D_{ta}, D_{tb}, \tau)\,.$$

C Recommended Empirical Benchmark Matrix

Model	Input	Purpose
Baseline	Returns	Raw predictive benchmark
Volatility	Realized volatility	Regime benchmark
Correlation	Return covariance	Cross-asset benchmark
PCA	Factor concentration	Common-factor benchmark
Cointegration	Residuals	Relative-value benchmark
Technical	RSI / MA / channels	Trading benchmark
Geometry	D_T	Structural benchmark
Full model	$\mathbf{S}_t$	Proposed framework

D Final Research Criterion

The framework should ultimately be evaluated according to

$$\boxed{\begin{array}{c}\text{Identification} + \text{Out-of-Sample Stability} + \text{Incremental Information} \\ + \text{Economic Significance} + \text{Replicability.}\end{array}}$$

Mathematical sophistication alone is insufficient. The central scientific test remains

$$I\,(\mathbf{S}_t; \text{future market state}) > I\,(\mathbf{B}_t; \text{future market state})\,, \tag{D.1}$$

where $\mathbf{B}_t$ denotes an appropriate benchmark state vector. Only if the geometric state provides statistically *and* economically significant incremental information should the Teichmüller component be considered empirically meaningful.

E Research-Status Matrix

Every component of the framework carries an explicit status, so that established mathematics is never silently blended with untested proposals.

Component	Status	Requirement
Quasiconformal / Teichmüller mathematics	Established	Standard theory [1, 2, 11]
Geometry in finance	Established	Literature comparison [13, 15, 16]
Teichmüller concepts in financial time series	Prior art	Cite and delimit [12]
Market tensor $P(t, a, b)$	Proposed	Formalize and test
Multi-venue market surface	Proposed	Synchronized data
Beltrami market field	Proposed	Numerical validation (Section 41.5)
Teichmüller-type deformation score	Proposed	Synthetic + empirical validation
Market compensator	Hypothesis	Test as representation operator
Trading edge	Unknown	Strict out-of-sample testing

F Open Mathematical Questions

1. What is the correct intrinsic market surface associated with $P(t, a, b)$?
2. What equivalence relation should define a financial Teichmüller space?
3. Which transformations are coordinate changes, and which are economically meaningful deformations?
4. Can a discrete Beltrami field be defined directly on the market tensor, without an arbitrary two-dimensional embedding?
5. Is there a natural graph- or discrete-Teichmüller formulation for cross-venue market data?
6. Can FX triangular closure be incorporated as a geometric constraint rather than a separate residual statistic?
7. Is there a useful analogue of a mapping-class or Veech-group action for recurring market geometries?
8. Can the geometry distinguish information arrival from microstructure noise?

Question 4 is the sharpest: an affirmative answer would remove the embedding choice of Section 17 — and with it the largest single source of arbitrariness in the construction.

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